Sampling

①

Sampling

Date

Ex population Consist of the value 0, 2, 4, 6, 8

a) Draw all possible Samples of size 3 with out Replacent (W.O.R)

Sol. ~~$N = 5$, $n = 2$~~

~~$^{N}C_{n} = {}^{5}C_{2} = \frac{5!}{2!(5-2)!} = 10$ Samples~~

Sol $N = 5$, $n = 3$

$^{N}C_{n} = {}^{5}C_{3} = \frac{5!}{3!(5-3)!} = \frac{5!}{3!\,2!}$

$^{5}C_{3} = \frac{5!}{3!\,2!} = \frac{5 \times 4 \times 3!}{3! \times 2!}$

$^{5}C_{3} = \frac{5 \times 4}{2} = 10$ Samples

Samples	$\bar{X} = a+b+c/3$
(0, 2, 4)	2
(0, 2, 6)	2.7
(0, 2, 8)	3.3
(0, 4, 6)	3.3
(0, 4, 8)	4
(0, 6, 8)	4.7
(2, 4, 6)	4
(2, 4, 8)	4.7
(2, 6, 8)	5.3
(2, 6, 8)	6

Population Mean μ

$\mu = 0 + 2 + 6 + 4 + 8 / 5$

$\mu = 4$

mean of all $\bar{x} = \frac{40}{10} = 4$

So that it is verified

$\mu = \bar{x}$

Sampling

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Date

Draw all possible samples of size 3 without Replacement (W.O.R)

0, 3, 12, 18, 30

i) Show that $\bar{x} = \mu$

ii) Find $V(\bar{x}) = \frac{\sigma^2}{n} * \frac{N-n}{N-1}$

Sol. $N = 5$, $n = 3$

$${}^5N_3 = \frac{5!}{3!(5-3)!} = \frac{5\times4\times3!}{3!\,2!} = 10$$

Samples	$\bar{x}$	$\bar{x}^2$
(0, 3, 12)	5	25
(0, 3, 18)	7	49
(0, 3, 30)	11	121
(0, 12, 18)	10	100
(0, 12, 30)	14	196
(0, 18, 30)	16	256
(0, 12, 30)	11	121
(3, 12, 18)	15	225
(3, 18, 30)	17	289
(12, 18, 30)	20	400
Total	126	1782

Mean of $\bar{x} = \frac{126}{10} = 12.6$

$\mu = 0+3+12+18+30/5$

$\mu = 12.6$

Hence $E(\bar{x}) = \mu$

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$$V(\bar{x}) = \frac{\sum x^2}{n} - (\sum x/n)^2$$

$$= \frac{1782}{10} - (126/10)^2$$

$$= 178.2 - 158.76$$

$$V(\bar{x}) = 19.44$$

Now Find σ^2

$$\sigma^2 = \frac{\sum x^2}{N} - (\sum N/N)^2$$

$$= \frac{1377}{5} - (63/5)^2$$

$$= 116.64$$

Now

$$\frac{\sigma^2}{n} \cdot \frac{N-n}{N-1} = \frac{116.64}{3} \times \frac{5-3}{5-1} = 19.44$$

$$V(\bar{x}) = \frac{\sigma^2}{n} \times \frac{N-n}{N-1}$$

Sampling

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Date

A Population Consists of Five 1, 3, 4, 7, 11

a) List all possible Sample of size 2 w.o.R.

b) Verify $E(\bar{x}) = \mu$ and

$$SE \text{ of } \bar{x} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

Sol. $N = 5, \quad n = 2 \quad {}^5C_2 = \frac{5!}{3!\,3!}$

$${}^5C_2 = \frac{5!}{2! \times 3!} = \frac{5 \times 4 \times 3!}{2!\,3!} = 10$$

Sample	$\bar{x}$	$\bar{x}^2$
(1, 3)	2	4
(1, 4)	2.5	6.25
(1, 7)	4.0	16
(1, 11)	6.0	36
(3, 4)	3.5	12.25
(3, 7)	5	25
(3, 11)	7	49
(4, 7)	5.5	30.25
(4, 11)	7.5	56.25
(7, 11)	9	81
	52	316

$E(\bar{x}) = 52/10 = 5.2$

population Mean $= \mu = \frac{1+3+7+4+11}{5} = 5.2$

Hence. $E(\bar{x}) = \mu$ verified.

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Now SE of $\bar{x} = \sqrt{V(\bar{x})}$

$$V(\bar{x}) = \frac{\Sigma \bar{x}^2}{n} - (\Sigma \bar{x}/n)^2$$

x	x^2
1	1
3	9
4	16
7	49
11	121
26	196

$$= \frac{316}{10} - (52/10)^2 = 4.56$$

$S.E$ of $\bar{x} = \sqrt{V(\bar{x})} = \sqrt{4.56} = 2.14$

Now

$$\sigma = \sqrt{\frac{\Sigma x^2}{N} - \left(\frac{\Sigma x}{N}\right)^2}$$

$$= \sqrt{\frac{196}{5} - (26/5)^2} = 3.4871$$

then

$$\frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}} = \frac{3.4871}{\sqrt{2}} \cdot \sqrt{\frac{5-2}{5-1}}$$

$$= 2.14$$

Hence. It is also verified that

$$SE \text{ of } \bar{x} = \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$$

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Date

Draw all possible sample of size 2 with replacement: 2, 0, 8, 6

Show that $E(\bar{x}) = \mu$.

Sol. Samples with Replacement $= (n)^N = (2)^4 = 16$

Samples	$\bar{x}$
(2,2)	2
(2,0)	1
(2,8)	5
(2,6)	4
(0,2)	1
(0,0)	0
(0,8)	4
(0,6)	3
(8,2)	5
(8,0)	4
(8,8)	8
(8,6)	7
(6,2) ~~(8,2)~~	4
(6,0)	3
(6,8)	7
(6,6)	6
	64

$$E(\bar{x}) = \frac{64}{16} = 4$$

$$\mu = 2+0+8+6/4 = 16/4 = 4$$

Hence Verified $E(\bar{x}) = \mu$.

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(9)

Date

Draw all possible samples of size 2 with Replacement from the population 9, 11, 15, 21

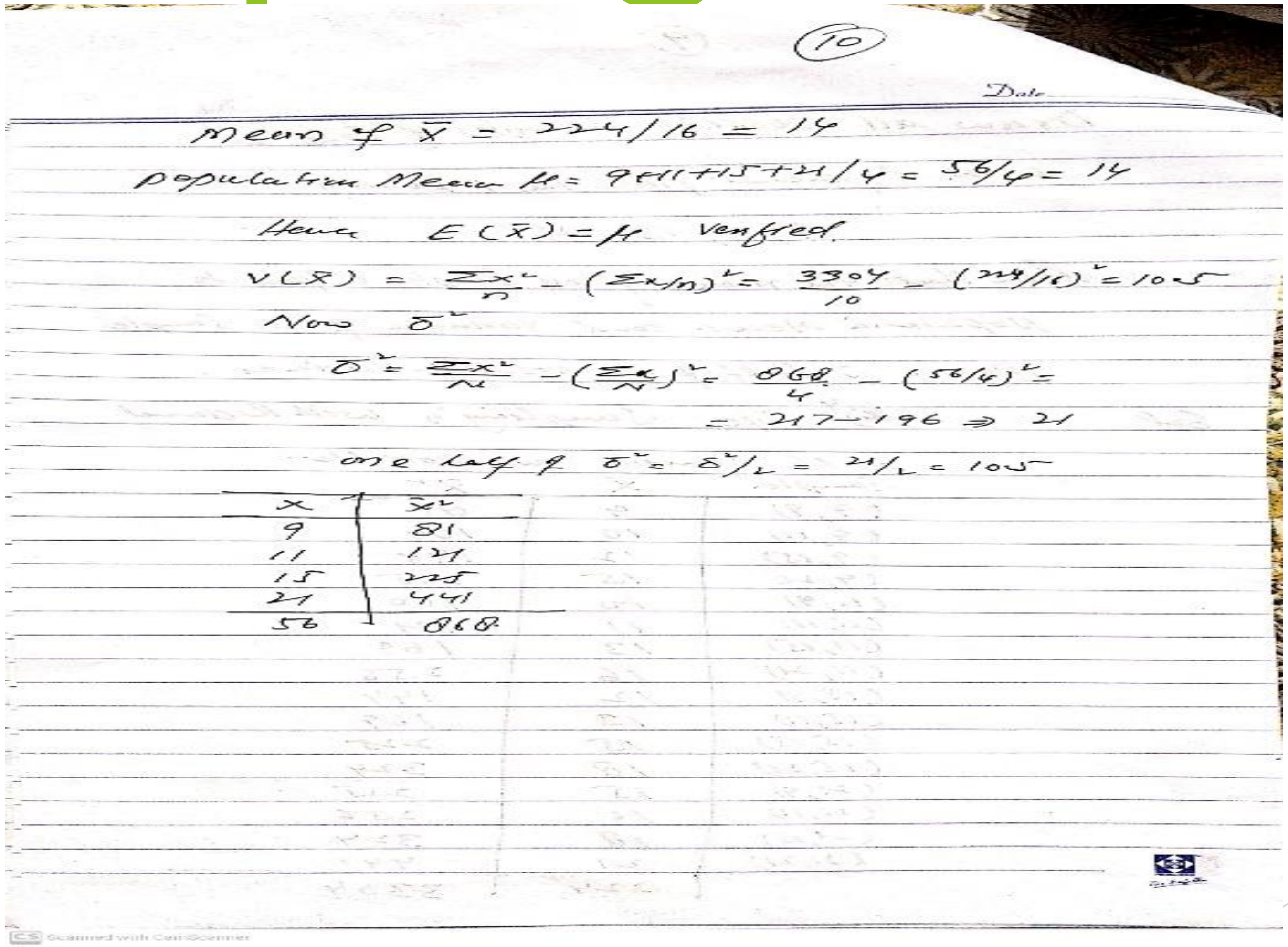
Show that the mean of all samples is equal to population mean. and variance of all sample means is half of the population variance.

Sol $(4)^2$ b/c sampling is with Replacement.

= 16 sample are possible.

Sample	$\bar{x}$	$\bar{x}^2$
(9, 9)	9	81
(9, 11)	10	100
(9, 15)	12	144
(9, 21)	15	225
(11, 9)	10	100
(11, 11)	11	121
(11, 15)	13	169
(11, 21)	16	256
(15, 9)	12	144
(15, 11)	13	169
(15, 15)	15	225
(15, 21)	18	324
(21, 9)	15	225
(21, 11)	16	256
(21, 15)	18	324
(21, 21)	21	441
	224	3304

Sampling



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Mean of $\bar{x} = 224/16 = 14$

population Mean $\mu = 9+11+15+21/4 = 56/4 = 14$

Hence $E(\bar{x}) = \mu$ verified.

$$V(\bar{x}) = \frac{\Sigma x^2}{n} - (\Sigma x/n)^2 = \frac{3304}{10} - (224/16)^2 = 10.5$$

Now σ^2

$$\sigma^2 = \frac{\Sigma x^2}{N} - \left(\frac{\Sigma x}{N}\right)^2 = \frac{868}{4} - (56/4)^2 =$$

$$= 217 - 196 \Rightarrow 21$$

one half of $\sigma^2 = \sigma^2/2 = 21/2 = 10.5$

x	x^2
9	81
11	121
15	225
21	441
56	868

Sampling

(11)

Date

Ex. Draw all possible samples of size 3 with Replacement from the population 24, 8. Show that $E(\bar{x}) = \mu$.

Sample variance = population variance.

Sol. $(N)^n = (2)^3 = 8$

Samples	$\bar{x}$	$s^2 = n\sum x^2 - (\sum x)^2 / n(n-1)$
(24, 24, 24)	24	0
(8, 24, 24)	18.7	85.33
(24, 8, 24)	18.7	85.33
(24, 24, 8)	18.7	85.33
(24, 8, 8)	13.3	85.33
(8, 24, 8)	13.3	85.33
(8, 8, 24)	13.3	85.33
(8, 8, 8)	8	0
	128	5119.8

Mean of $\bar{x} = 128/8 = 16$

population Mean $= \mu = 24+8/2 = 16$

Hence

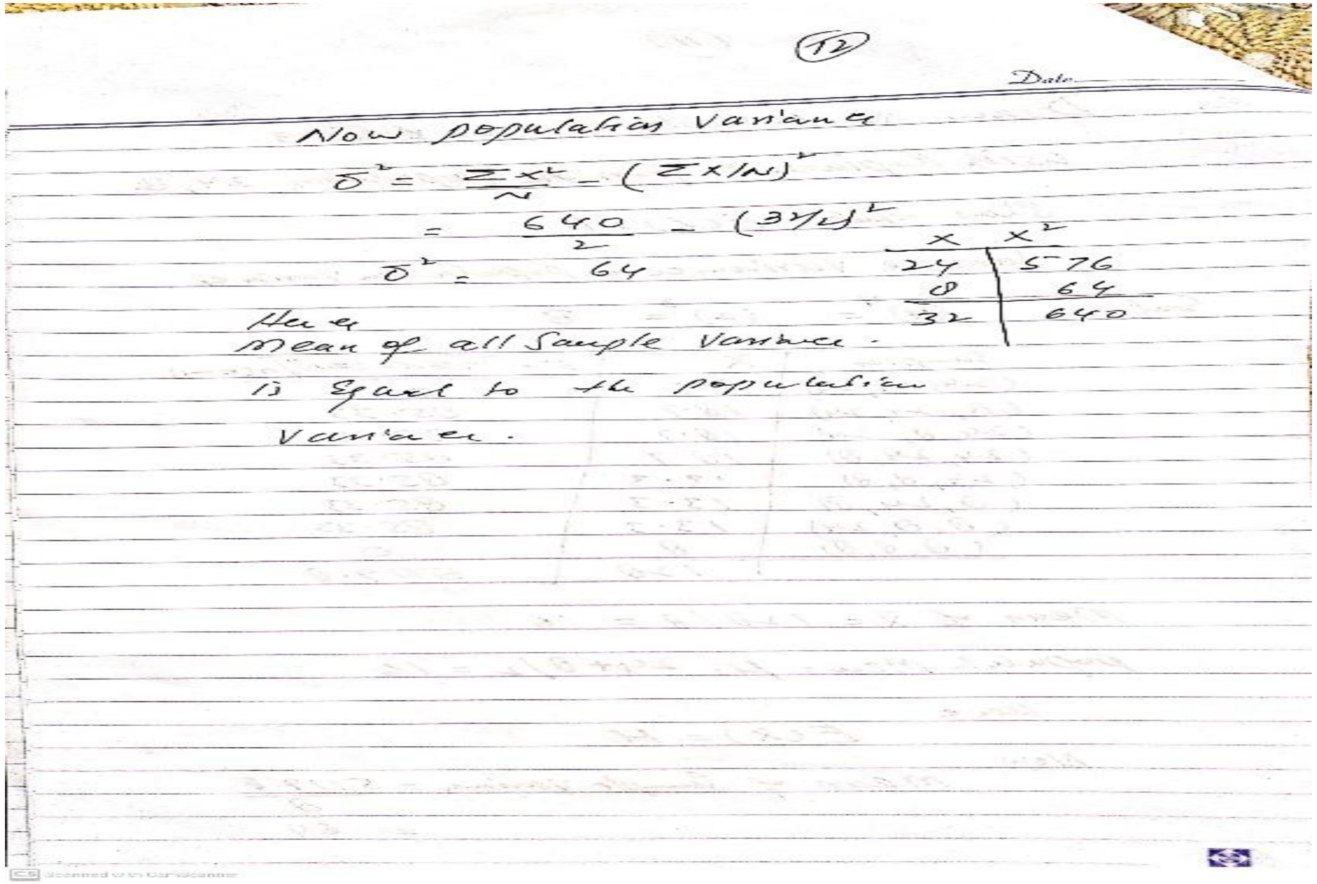
$E(\bar{x}) = \mu$.

Now

Mean of Sample variance $= \frac{5119.8}{8}$

$= 64$

Sampling



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Now population variance

$$\sigma^2 = \frac{\Sigma x^2}{N} - (\Sigma x/N)^2$$

$$= \frac{640}{2} - (32/2)^2$$

$$\sigma^2 = 64$$

x	x^2
24	576
8	64
32	640

Hence

Mean of all sample variance.

is equal to the population

variance.

Sampling

(13)

If a population is given as (2, 3, 6, 8, 11). After Selecting all possible Sample of size 2 with Replacement Find $E(\bar{x} - \mu)$ and Standard Error of $\bar{x}$.

Sol: $(N)^n = (5)^2 = 25$ possible Samples

Sample	$\bar{x}$	$\bar{x} - 6$	Sample	$\bar{x}$	$\bar{x} - 6$
(2, 2)	2	-4	(11, 3)	7	1
(2, 3)	2.5	-3.5	(11, 6)	8.5	2.5
(2, 6)	4	-2	(11, 8)	9.5	3.5
(2, 8)	5	-1	(11, 11)	11	5
(2, 11)	6.5	0.5		150	0
(3, 2)	2.5	-3.5			
(3, 3)	3	-3			
(3, 6)	4.5	-1.5			
(3, 8)	5.5	-0.5			
(3, 11)	7	1			
(6, 2)	4	-2			
(6, 3)	4.5	-1.5			
(6, 6)	6	0			
(6, 8)	7	1			
(6, 11)	8.5	2.5			
(8, 2)	5	-1			
(8, 3)	5.5	-0.5			
(8, 6)	7	1			
(8, 8)	8	2			
(8, 11)	9.5	3.5			
(11, 2)	6.5	0.5			

Mean of $\bar{x} = \frac{150}{5} = 6$

$\mu = \frac{2+3+6+8+11}{5} = 6$

$E(\bar{x}) = \mu$

$6 = 6$

Sampling

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SE of $\bar{x} = \sqrt{V(\bar{x})}$

$V(\bar{x}) = \sigma^2/n$

SE of $\bar{x} = \sqrt{V(\bar{x})} = \sqrt{\sigma^2/n} = \sigma/\sqrt{n}$.

$\sigma^2 = \sqrt{\frac{\Sigma x^2}{N} - \left(\frac{\Sigma x}{N}\right)^2}$

$= \sqrt{\frac{234}{5} - (30/5)^2} = 3.2863$

SE of $\bar{x} = \frac{3.2863}{\sqrt{2}} = 2.32$

$E(\bar{x} - \mu) = \frac{\Sigma(\bar{x} - 6)}{n} = 0/25 = 0$

x	x^2
2	4
3	9
6	36
8	64
11	121
30	234

Sampling

(15)

Date

If the Sample Size is 36 and the S.E of the Mean is 2 what should be the Size of the Sample if the SE is to be reduced to 1.2 ?

$n = 36, \quad SE = \sigma/\sqrt{n} = 2$

$$\frac{\sigma}{\sqrt{n}} = 2$$

$$\sigma = 2\sqrt{n} = 2(\sqrt{36}) = 12$$

if $SE = 1.2$

$$\frac{\sigma}{\sqrt{n}} = 1.2$$

$$\sigma = 1.2\sqrt{n}.$$

$$\frac{\sigma}{1.2} = \sqrt{n} \qquad S.B.S$$

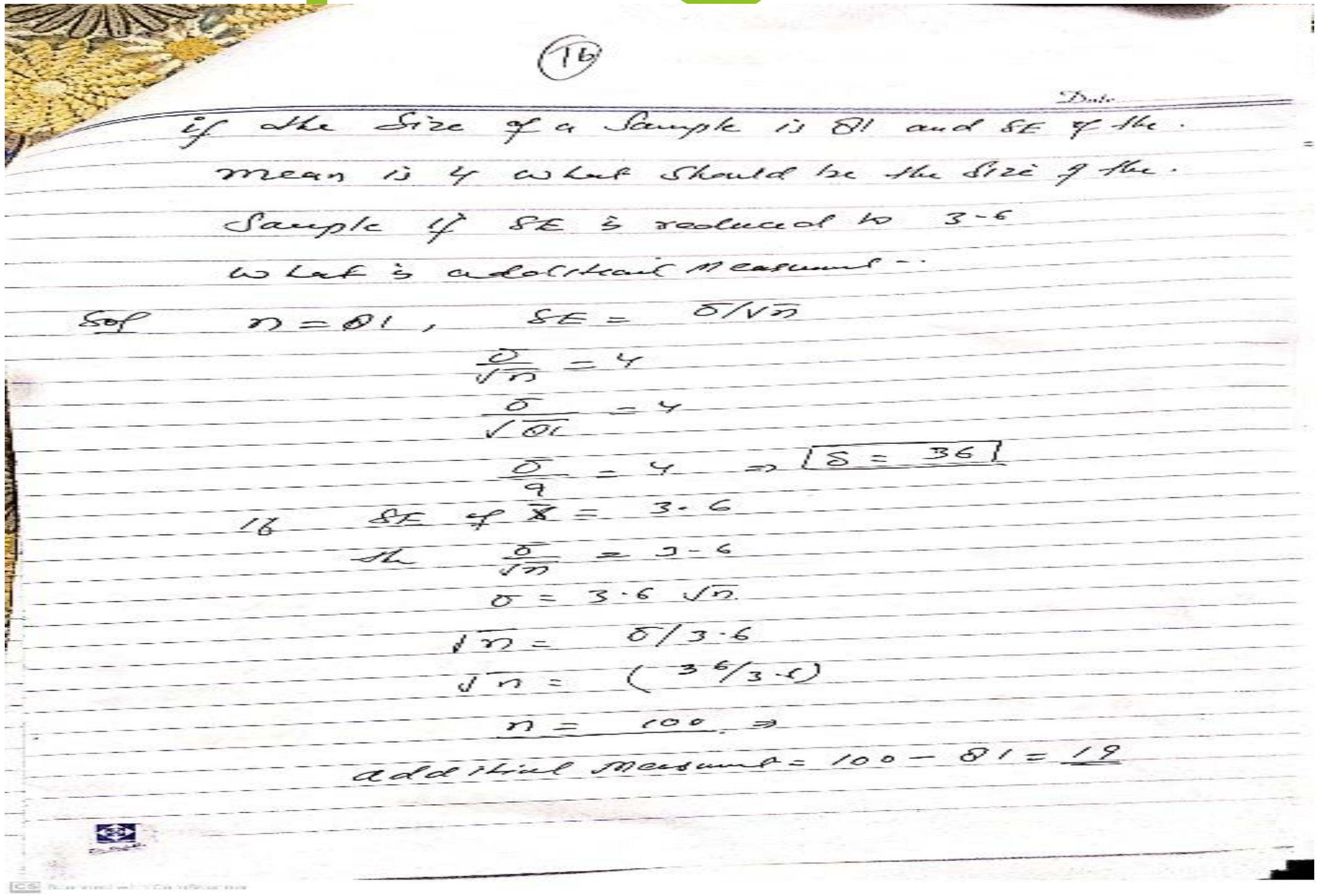
$$\frac{\sigma^2}{(1.2)^2} = n$$

$$n = (\sigma/1.2)^2$$

$$n = (12/1.2)^2 = (120/12)^2.$$

$$\boxed{n = 100}$$

Sampling



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if the size of a Sample is 81 and SE of the mean is 4 what should be the size of the Sample if SE is reduced to 3.6

what is additional measurement.

Sol $n = 81$, $SE = \sigma/\sqrt{n}$

$$\frac{\sigma}{\sqrt{n}} = 4$$

$$\frac{\sigma}{\sqrt{81}} = 4$$

$$\frac{\sigma}{9} = 4 \Rightarrow \boxed{\sigma = 36}$$

If SE of $\bar{x}$ = 3.6

then $\frac{\sigma}{\sqrt{n}} = 3.6$

$\sigma = 3.6\sqrt{n}$

$\sqrt{n} = \sigma/3.6$

$\sqrt{n} = (36/3.6)$

$n = 100 \Rightarrow$

additional measurement = 100 − 81 = 19

Sampling

Chapter - 11

Sampling

Introduction

The process of selecting a small part from a large collection, such that the selected part will show all the characteristics of the large collection is called sampling. The small part which is taken out from a large collection is called "Sample" and the large collection is called "Population".

Sampling is quite often used in our day-to-day practical life. For example, in a shop we assess the quality of rice, sugar and any other commodity by taking a handful of it from the large and then decide to purchase it or not. A house wife normally tests the cooked products to find if they are properly cooked and contain the proper quantity of salt.

There are so many techniques to draw a sample from a population. Here we shall discus only the techniques of Simple Random Sampling.

Before introducing the simple random sampling, some of the important terms and the basic concepts are defined first.

Population

A complete collection of individuals, objects or measurements under consideration in a statistical study will be called a population.

OR

A population may be defined as the large collection of similar units. The number of units in the population (i.e. size of a population) is always denoted by N.

For example, a population may be of heights of all students of a college, a population may be of all books in a library, a population may be of all persons using a particular medicine, a population may be of prices of a commodity over a period of time, etc.

Sampling

A population may be finite or infinite depending upon whether it contains a countable number of units or an uncountable number of units.

Sample

A small part of a population is called sample. The number of units in the sample (i.e. size of a sample) is always denoted by n.

Parameter

Any quantity calculated from a population is called a parameter. For example, population mean, population variance, etc. are therefore parameters. The population mean is denoted by μ and the population variance is denoted by σ^2.

Statistic

Any quantity calculated from a sample is called a statistic. For example, sample mean, sample variance, etc. are therefore statistics. The sample mean is denoted by $\bar{x}$ and the sample variance is denoted by s^2

Sampling with and without replacement

Sampling is said to be with replacement when we draw a unit from a finite population and return it to the population before the next unit is drawn. In this case each unit can be drawn more than once and the probability of drawing of each unit remains constant throughout the sampling procedure.

Sampling is said to be without replacement, if we do not return the selected unit to the population and draw the next unit. In this case each unit cannot be drawn more than once and the probability of drawing of each unit changes throughout the sampling procedure.

Census and Sample Survey

Census or complete enumeration means to get the information about each and every unit in the population.

A sample survey is a technique of getting information about the characteristics of the population by studying only a part (i.e by studying a sample) of the population.

Sampling

sample survey.

Simple Random Sampling

This is the simplest and the easiest method of drawing a sample from a population. According to this method each and every unit in the population has an equal chance of being included in the sample and also each possible sample of the same size has an equal probability of being chosen.

Suppose, there is a population of N units and we want to draw a sample of n units, then the possible number of samples in case of sampling without replacement will be ${}^{N}C_{n} = \frac{N!}{n!\,(N-n)!}$ and in case of sampling with replacement the possible number of samples will be N^n

As an illustration, suppose there is a population of 4 persons, identified as A, B, C and D and we want to select a random sample of 2 persons.

Therefore $N = 4$, $n = 2$ and there will

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is called lottery method. This method is not convenient when we have a large population, so we mostly apply computer programs.

Sampling Distributions

Consider all possible samples of size n which can be drawn from a given population (either with or without replacement). For each sample we can compute a statistic, such as the mean, variance, etc. Which will vary from sample to sample. In this way we obtain a distribution of the statistic which is called its sampling distribution. Therefore, the sampling distributions may be of mean, variance, etc.

Sampling Distribution of the mean ($\bar{x}$)

If we draw all possible samples each of size n from a finite population of N units with mean μ and variance σ^2. Then we compute mean of every sample i.e. $\bar{x}$. Therefore, the statistic ($\bar{x}$) is now a random variable and form a probability distribution. This distribution is called sampling distribution of mean.

The sampling distribution of mean has the following properties in case of sampling without replacement and in case of sampling with replacement.

Properties in case of Sampling without replacement

1. Mean of all sample means is equal to the population mean i.e $E(\bar{x}) = \mu$ or $\bar{x}$ is said to be an unbiased estimator of μ.

2. $V(\bar{x}) = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1}$ where σ^2 is the population variance.

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Properties in case of Sampling with replacement

1. Mean of all sample means is equal to the population mean i.e $E(\bar{x}) = \mu$ or $\bar{x}$ is said to be an unbiased estimator of μ.

2. $V(\bar{x}) = \frac{\sigma^2}{n}$ where σ^2 is the population variance.

Standard Error

The standard deviation of sampling distribution of $\bar{x}$ is called standard error of $\bar{x}$.

Thus S.E. of $(\bar{x}) = \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$ in case of without replacement

and S.E. of $(\bar{x}) = \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ in case of with replacement

Example 1

A population consists of five numbers 0, 2, 4, 6, 8

a) List all possible samples of size 2 that can be drawn from this population without replacement

b) Find mean of each sample

c) Construct sampling distribution of $\bar{x}$.

d) Verify that mean of all sample means is equal to population mean.

Solution:

Since N = 5 and n = 2 and the sampling is don without replacement, then all possible samples = ${}^{5}C_{2} = \frac{5!}{2!\,(5-2)!} = 10$ as shown below;

Sampling

Sample No.	All Possible Samples	Sample Mean $\bar{x}$
1	(0, 2)	1
2	(0, 4)	2
3	(0, 6)	3
4	(0, 8)	4
5	(2, 4)	3
6	(2, 6)	4
7	(2, 8)	5
8	(4, 6)	5
9	(4, 8)	6
10	(6, 8)	7
Total		40

Sampling Distribution of $\bar{x}$ is

$\bar{x}$	1	2	3	4	5	6	7
f	1	1	2	2	2	1	1

Now find mean of all means i.e. mean of sampling distribution of $\bar{x}$ is computed as

$\bar{x}$	f	$f\bar{x}$
1	1	1
2	1	2
3	2	6
4	2	8
5	2	10
6	1	6
7	1	7
Total	10	40

The Mean of $\bar{x} = E(\bar{x}) = \frac{\sum f\bar{x}}{\sum f} = \frac{40}{10} = 4$

and since, population mean $\mu = \frac{0+2+4+6+8}{5} = 4$

Therefore, it is verified that , mean of all sample means is equal to population mean.

Sampling

Example 2

A population consists of five numbers 0, 3, 6, 9, 12

a) List all possible samples of size 3 that can be draw from this population without replacement

b) Verify that, Mean of $\bar{x} = E(\bar{x}) = \mu$ and $V(\bar{x}) = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1}$

Solution:

Since N = 5 and n = 3 and sampling is done without replacement. Then all possible samples $= {}^5C_3 = \frac{5!}{3!\,(5-3)!} = 10$

Population mean and variance are computed as

$$\mu = \frac{\sum x}{N} = \frac{30}{5} = 6$$

$$\sigma^2 = \frac{\sum x^2}{N} - \left(\frac{\sum x}{N}\right)^2$$

$$\sigma^2 = \frac{270}{5} - \left(\frac{30}{5}\right)^2 = 18$$

	x	x^2
	0	0
	3	9
	6	36
	9	81
	12	144
Total	30	270

The mean and variance of $\bar{x}$ are computed as

All Possible Samples	Sample Mean $\bar{x}$	$\bar{x}^2$
(0, 3, 6)	3	9
(0, 3, 9)	4	16
(0, 3, 12)	5	25
(0, 6, 9)	5	25
(0, 6, 12)	6	36
(0, 9, 12)	7	49
(3, 6, 9)	6	36
(3, 6, 12)	7	49
(6, 9, 12)	9	81
(3, 9, 12)	8	64
Total	60	390

Sampling

Mean of $\bar{x} = E(\bar{x}) = \frac{\sum \bar{x}}{m} = \frac{60}{10} = 6 = \mu$, where m = No. of samples

Therefore, it is verified that $E(\bar{x}) = \mu$

Now $V(\bar{x}) = \frac{\sum \bar{x}^2}{m} - \left(\frac{\sum \bar{x}}{m}\right)^2$ where m = numbers of samples

$$V(\bar{x}) = \frac{390}{10} - \left(\frac{60}{10}\right)^2 = 3$$

Also $V(\bar{x}) = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1} = \frac{18}{3} \cdot \frac{5-3}{5-1} = 3$

Hence it is also be verified that $V(\bar{x}) = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1}$

Example 5

Draw all possible samples each of size 2 from the population 2, 4, 6 and 8 using samplling with replacement. Find mean of each sample and verify that

(i) Mean of $\bar{x} = \mu$ and (ii) $V(\bar{x}) = \frac{\sigma^2}{2}$

Solution:

Firstly, we compute Population mean and varaince

$$\mu = \frac{\sum x}{N} = \frac{20}{4} = 5$$

$$\sigma^2 = \frac{\sum x^2}{N} - \left(\frac{\sum x}{N}\right)^2$$

$$\sigma^2 = \frac{120}{4} - \left(\frac{20}{4}\right)^2 = 5$$

	x	x^2
	2	4
	4	16
	6	36
	8	64
Total	20	120

Since N = 4 and n = 2 then, possible samples $= N^n = 4^2 = 16$

The possible samples and their means are computed as

Sampling

Possible Samples	Sample Mean $\bar{x}$	$\bar{x}^2$
(2, 2)	2	4
(2, 4)	3	9
(2, 6)	4	16
(2, 8)	5	25
(4, 2)	3	9
(4, 4)	4	16
(4, 6)	5	25
(4, 8)	6	36
(6, 2)	4	16
(6, 4)	5	25
(6, 6)	6	36
(6, 8)	7	49
(8, 2)	5	25
(8, 4)	6	36
(8, 6)	7	49
(8, 8)	8	64
Total	80	440

Mean of $\bar{x} = \frac{\sum \bar{x}}{m} = \frac{80}{16} = 5 = \mu$. Therefore Mean of $(\bar{x}) = \mu$

Now $V(\bar{x}) = \frac{\sum \bar{x}^2}{m} - \left(\frac{\sum \bar{x}}{m}\right)^2$ where m = numbers of samples

$$V(\bar{x}) = \frac{440}{16} - \left(\frac{80}{16}\right)^2 = 2.5$$

Also $V(\bar{x}) = \frac{\sigma^2}{n} = \frac{5}{2} = 2.5$

Hence it is also be verified that $V(\bar{x}) = \frac{\sigma^2}{n}$

Example 4

Draw all possible samples each of size n = 3 from the population 2, 11 by using sampling with replacement. Find standard error of sampling distribution of $\bar{x}$.